ATMOS / EIP 2026 TECHNICAL PITCH

AI-assisted operations. Deterministic control.

Ensuring safe inventory decisions through a clear trust boundary between AI perception and inventory authority.

V3.5.3 TECHNICAL PROTOTYPE · SOLO FOUNDER BUILD · EGYPT

Critical inventory decisions are hard to inspect.

[SYNTHETIC / NOT PRODUCTION]

Unchecked Intake

AI-assisted data entry can propose plausible but unsafe values for expiry, batch, or quantity.

Siloed Controls

Information moves between invoices, messages, and legacy systems, losing the decision trail.

Containment Lag

When a recall occurs, identifying and blocking affected stock takes hours of manual reconstruction.

BUSINESS CONSEQUENCE

Human review alone is an alert, not a control.

- → Increased risk of dispensing expired batches
- → Delayed response to critical recalls
- → Opaque audit trail for compliance
- → Operational rework and wastage

AI may propose. Rules decide.

01 / PROBABILISTIC

AI Perception

Gemma extracts structured field ɔm raw invoice text. *(Untrusted Input)*

TRUST BOUNDARY

02 / DETERMINISTIC

Policy Gate

Strict validation of SKU, Expiry, Batch, and Roles before mutation.

03 / VERIFIABLE

Audit Evidence

SHA-256 hash linked ledger and state seal for every decision.

CORE INVARIANT

The model never writes inventory balances directly. AI proposes; the deterministic core executes.

What is proven today

[SYNTHETIC / NOT PRODUCTION]

2,000

WORKLOAD TRANSACTIONS

Deterministic stress test of inventory state transitions and receipt gates.

102

INJECTED SCENARIOS

Expired goods, batch mismatches, and unauthorized roles deliberately introduced.

100%

FAIL-CLOSED ACCURACY

Every unsafe scenario was identified and blocked by the deterministic core in this fixture.

DETERMINISTIC EVIDENCE: These figures come from a reproducible synthetic workshop fixture. They are not production measurements or capacity claims. CI validation includes normalized PostgreSQL, tenant isolation, and local OIDC protocol tests.

The first customer and the Pilot

Phase 1: Controlled Shadow Pilot

6–8 week deployment at one site. Anonymized data intake running in parallel to existing WMS/ERP. No operational authority granted.

Phase 2: Baseline & Evidence

Measuring trace-evidence completeness, exception handling accuracy, and manual rework cycles against the established baseline.

Phase 3: Documented Scale Decision

Review of Shadow results with Quality and Operations. Approval of limited enforcement for selected high-risk receipt gates.

TARGET CUSTOMER

Mid-sized pharma distributor or pharmacy chain with central warehouse operations.

A focused ask: fund validation

OBJECTIVE

Transition to Field Staging

Moving from synthetic local testing to a secure staging environment with real data-source integration.

PATH

Execute Controlled Pilot

Funding for one site deployment, secure cloud staging, real OIDC/Entra identity, and user testing.

PAYOFF

Documented Scale Evidence

A verifiable report on efficiency, safety, and adoption to decide on full-scale production rollout.

GATE Support validation and Pilot execution—not unbounded platform complexity. All field work is governed by an explicit Shadow-First policy.

MISSION STATEMENT

Safer to inspect.
Harder to bypass.

Next Milestone: Staging Validation · Nov 2026
Pilot Phase 1 · Dec 2026